



Northern University

Of Business and Technology Khulna

Course Title: Mobile Computing Lab

Course Code: CSE 4204

Frontend Progress

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Enterprise Nexus — Multi-Agent AI Business Automation OS with Security Intelligence

1. Project Title & Overview

Project Title: Enterprise Nexus — Multi-Agent AI Business Automation OS with Security Intelligence.

Enterprise Nexus is a multi-agent AI business automation platform that unifies core departmental functions — HR, Finance, Support, Analytics, and Executive intelligence — into a single, role-based web dashboard, with a dedicated Security Intelligence module for threat monitoring. This report documents the Week 07 milestone: transforming the approved Week 05 UI/UX designs into a functional Next.js frontend and integrating it with the REST APIs delivered in Week 06.

2. Team Information

Role	Name	Student ID
Team Leader & AI Integration Lead	Nazmus Sakib	11220320888
Backend Developer	Shoeb Shariar Mashuk	11220320878
Frontend Developer — 1	Most Sumiya Sanjida	11220320874
Database Manager & Frontend Developer — 2	Sabrina Ibrahim	11220320895

3. Frontend Technology Stack

The frontend is built on the Next.js App Router with TypeScript for type safety and a modular, scalable project structure.

Layer	Technology
Framework	Next.js 14 (App Router) + React 18
Language	TypeScript
Styling	Tailwind CSS 3
HTTP Client	Axios (with interceptors)
Data Fetching	SWR (stale-while-revalidate)
Forms & Validation	React Hook Form
Auth Sessions	JWT via httpOnly cookies (js-cookie)
Notifications / Feedback	react-hot-toast
Charts	Recharts
Icons	lucide-react

Layer	Technology
Monitoring	Sentry
Code Quality	ESLint + Prettier + Husky pre-commit

4. Completed Pages

Seventeen screens have been implemented across two route groups — a public authentication group and a protected dashboard group — closely following the Week 05 UI designs.

Authentication & Public Pages

Page	Route	Status
Landing / Home	/	Completed
Login	/login	Completed
Registration	/register	Completed
Forgot Password	/forgot-password	Completed
Reset Password	/reset-password	Completed

Protected Dashboard Pages

Page	Route	Status
Main Dashboard	/dashboard	Completed
HR Agent (list + detail)	/hr, /hr/[id]	Completed
Finance Agent (list + detail)	/finance, /finance/[id]	Completed
Support Agent (list + detail)	/support, /support/[id]	Completed
Analytics	/analytics	Completed
Executive Briefing	/executive	Completed
Security Intelligence	/security	Completed
Admin Panel	/admin	Completed
User Profile & Settings	/settings	Completed

5. Backend APIs Integrated

Data is loaded dynamically from the Week 06 REST backend through a centralized Axios client (src/lib/api.ts). No hardcoded or dummy data is used where an API is available. A response

interceptor transparently refreshes expired tokens and redirects to login on failure. The following API modules are wired to the UI:

Module	Key Endpoints	Frontend Usage
Auth	register, login, refresh, logout, me, forgot/reset-password	Login, register, session, profile
HR	reports (CRUD), stats, CV upload	HR agent screens
Finance	records (CRUD), anomalies, summary, invoice upload	Finance agent screens
Support	tickets (CRUD), escalate, resolve	Support desk screens
Analytics	kpi, stats, reports	Analytics dashboard
Executive	briefing, briefing PDF, generate	Executive page
Security	audit-logs, anomalies	Security module
Admin	users, roles, role/status update	Admin panel
Notifications	list, read, read-all	Top-bar notifications
Search	semantic / global search	Global search bar

6. Authentication Flow

A complete JWT-based authentication flow is implemented end to end:

1. **Registration & Login** — credentials are validated client-side, then posted to `/auth/register` and `/auth/login`; the backend issues JWT access and refresh tokens as `httpOnly` cookies.
2. **Session Management** — a React `AuthContext` calls `/auth/me` on load to restore the session and expose the current user and role to the app.
3. **Silent Refresh** — the Axios interceptor catches 401 responses, calls `/auth/refresh` once, and retries the original request automatically.
4. **Protected Routes** — the dashboard route-group layout guards every protected page, showing a loading state and redirecting unauthenticated users to `/login`.
5. **Logout** — `/auth/logout` clears the session cookies and returns the user to the login page.
6. **Unauthorized Handling** — role-based access is enforced (RBAC), and rate limiting on `login/reset` endpoints protects against brute-force attempts.

7. Form Validation

All input forms use React Hook Form with client-side rules and inline, user-friendly error messages:

- Required-field checks on all login, registration, and data-entry forms.
- Email format validation and password strength / confirmation rules.
- Invalid or incomplete submissions are blocked before reaching the API.

- Server-side validation errors are surfaced back to the user via toast notifications.

8. Responsive Design

The interface is built mobile-first with Tailwind's responsive utilities and verified across desktop, tablet, and mobile breakpoints:

- The sidebar collapses into an overlay drawer on mobile; a fixed top bar remains accessible.
- Content, grids, and charts reflow and re-align at md and lg breakpoints.
- Forms, buttons, and menus stay usable and touch-friendly on small screens.

9. User Experience Improvements

Reusable UI primitives provide consistent, professional feedback across the app:

- Loading indicators and skeleton screens during data fetches.
- Success and error toasts for every user action (react-hot-toast).
- Empty-state messages for lists with no data.
- Confirmation dialogs for destructive actions (delete, role changes).
- An error boundary and a guided welcome tour for first-time users.

10. Current Development Progress

Area	Progress	Notes
Project setup & structure	100%	Next.js App Router, TypeScript, Tailwind, linting
UI screens	100%	17 pages built to Week 05 designs
Backend API integration	95%	All core modules wired via Axios client
Authentication flow	100%	JWT, refresh, protected routes, RBAC
Form validation	100%	React Hook Form across all forms
Responsive design	90%	Desktop / tablet / mobile verified
UX polish	85%	Loading, toasts, empty states, dialogs
AI agent integration	In progress	Scheduled for Week 08 (Gemini, LangChain, CrewAI)

11. GitHub Repository

Repository: <https://github.com/Nazmussakib247/CSE4204-8B-T04-Enterprise-Nexus-Multi-Agent-AI-Business-Automation-OS-with-Security-Intelligence>

12. Screenshots

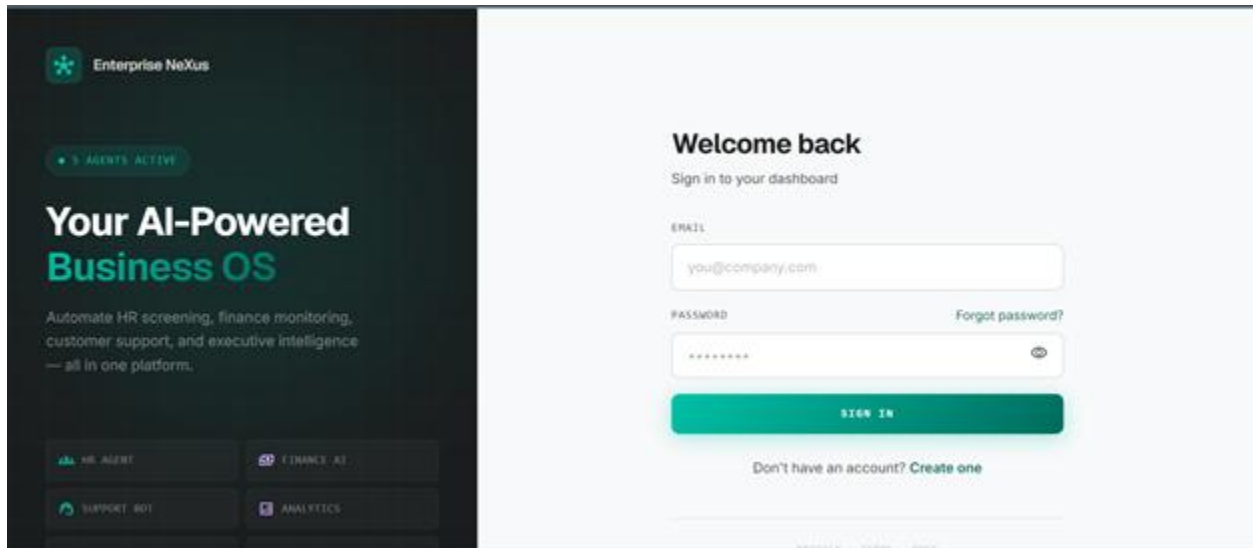


Figure 1: Login Page

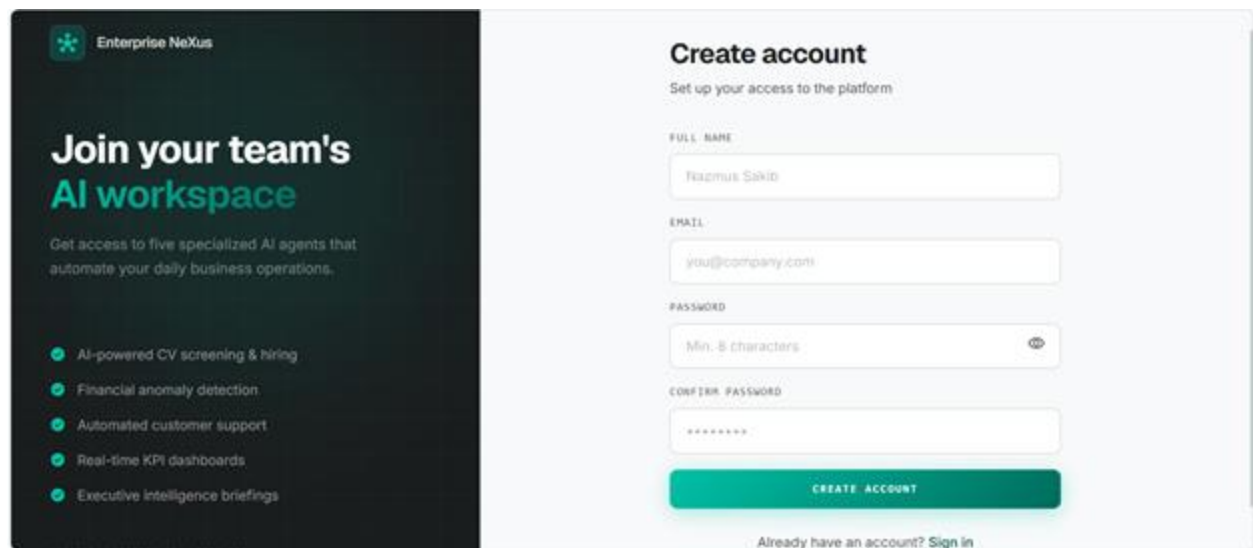


Figure 2: Registration Page

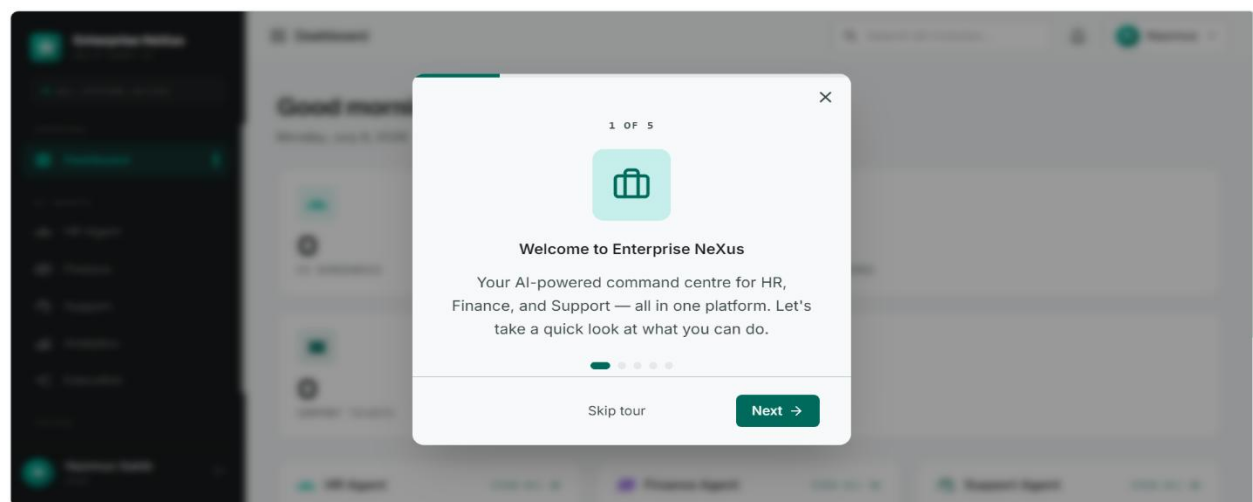


Figure 3: Welcome Tour

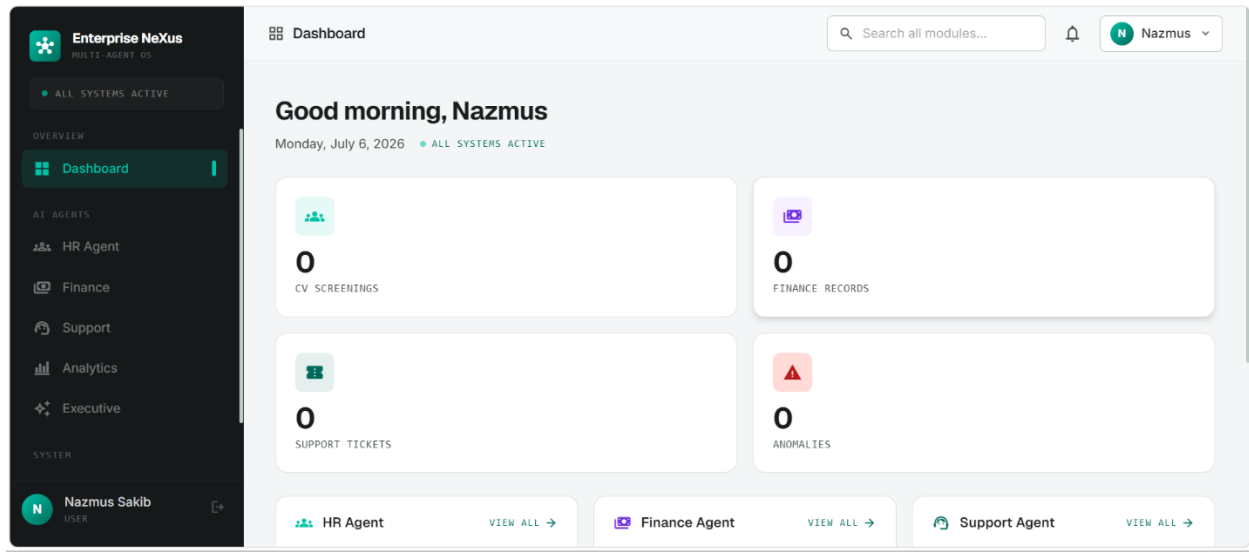


Figure 4: Dashboard

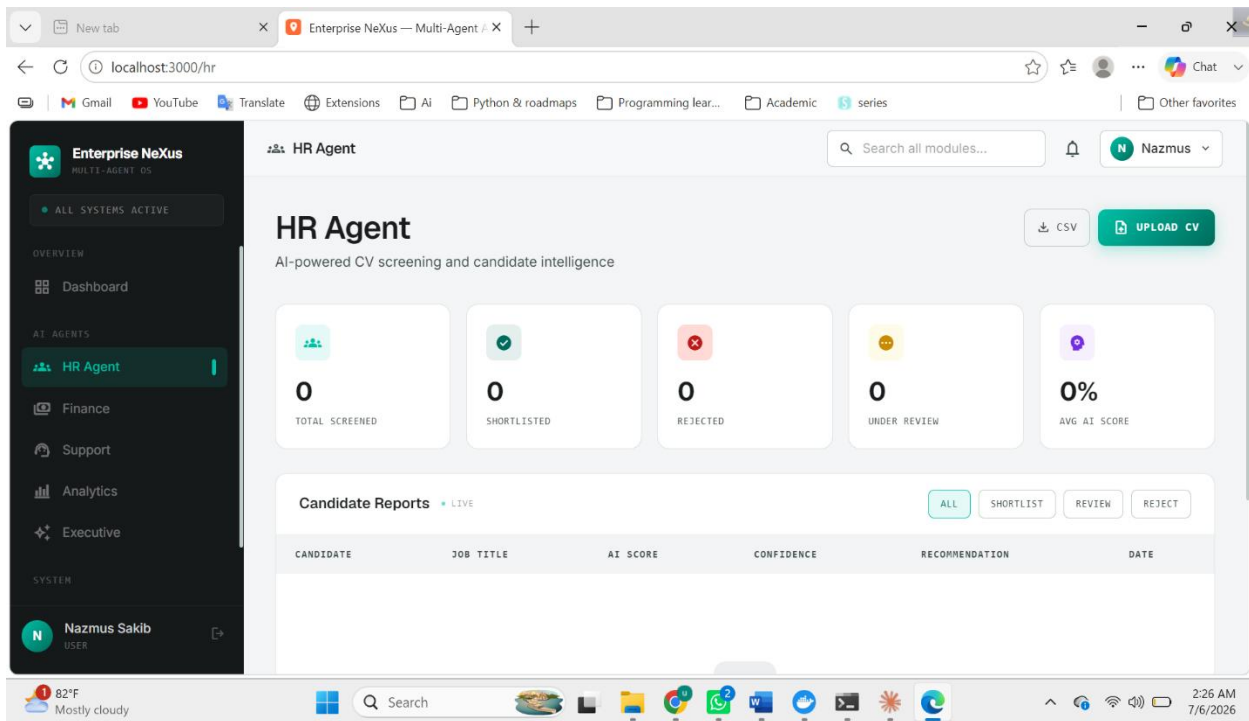


Figure 5: HR Agent

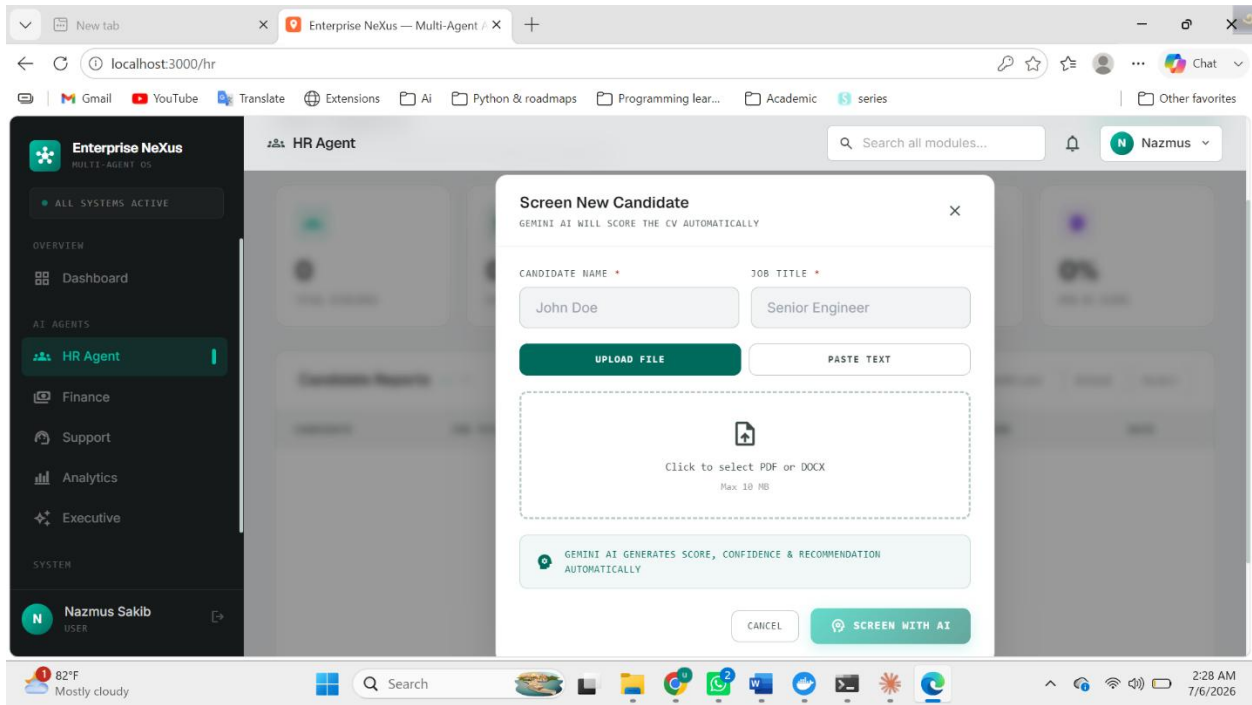


Figure 6: HR Agents/ CV Screening Function

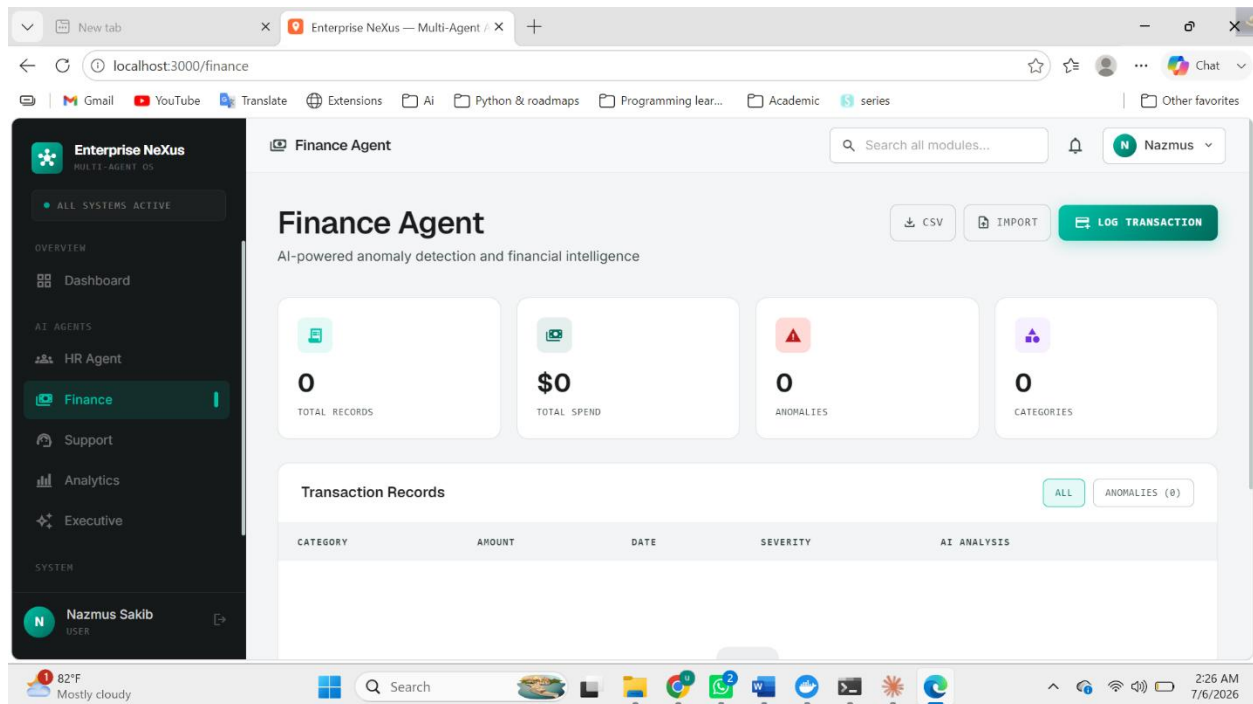


Figure 7: Finance Agents

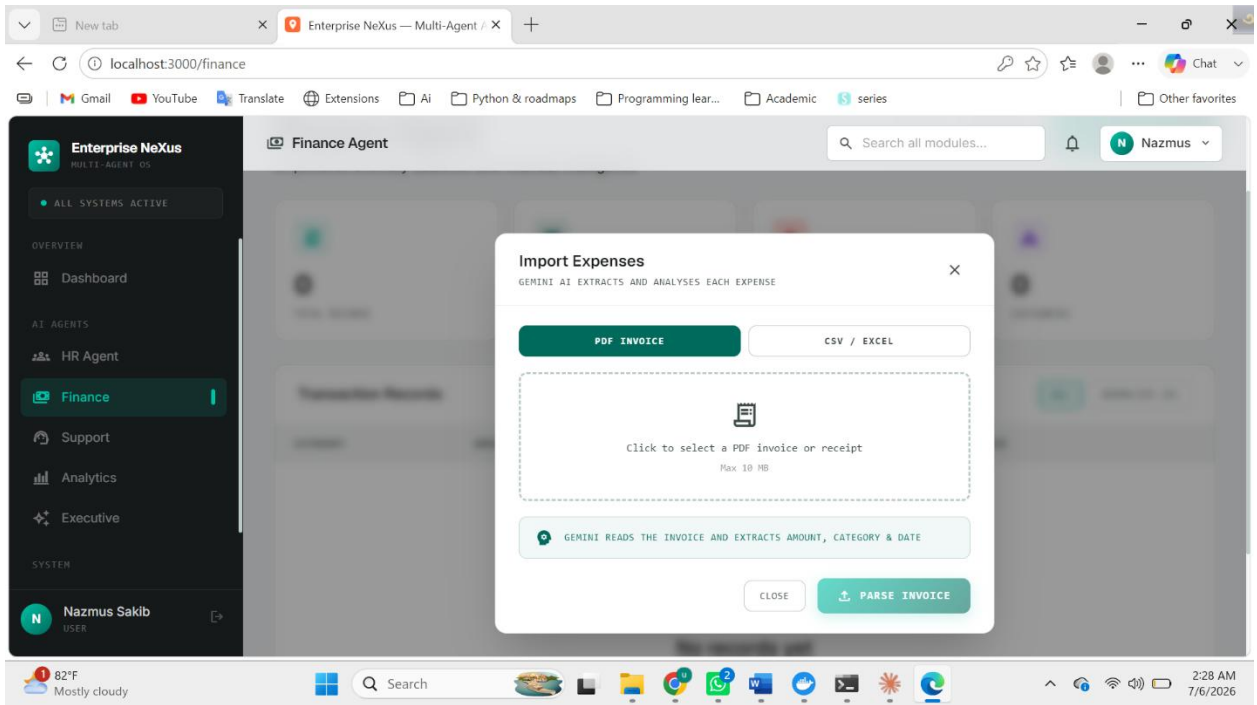


Figure 8: Finance agents / Invoice add-CSV-Excel

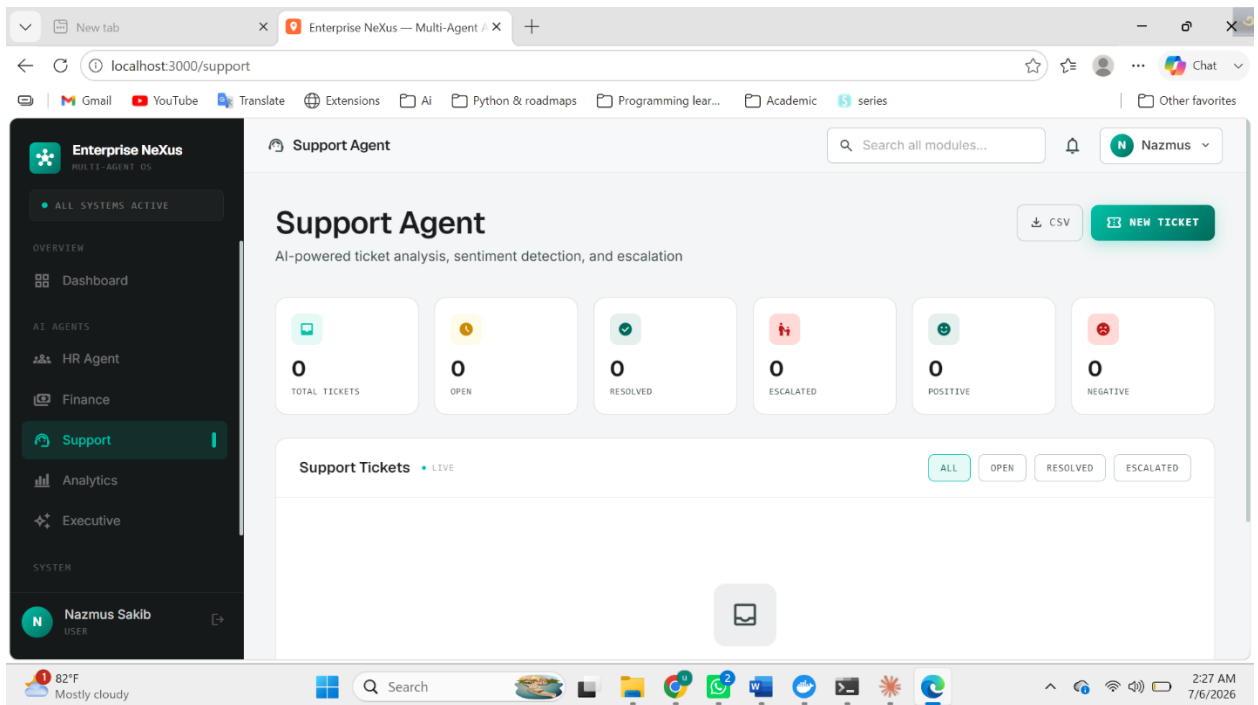


Figure 9: Support Agent

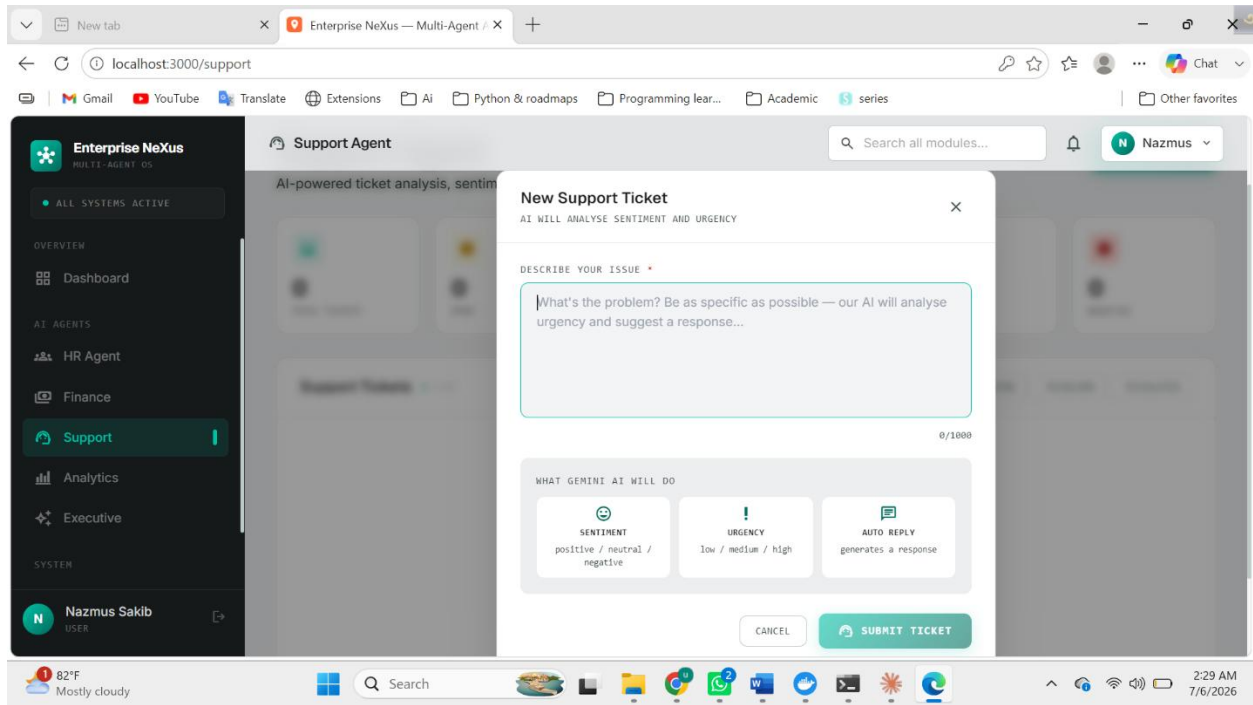


Figure 10: Support Agent / Ticket

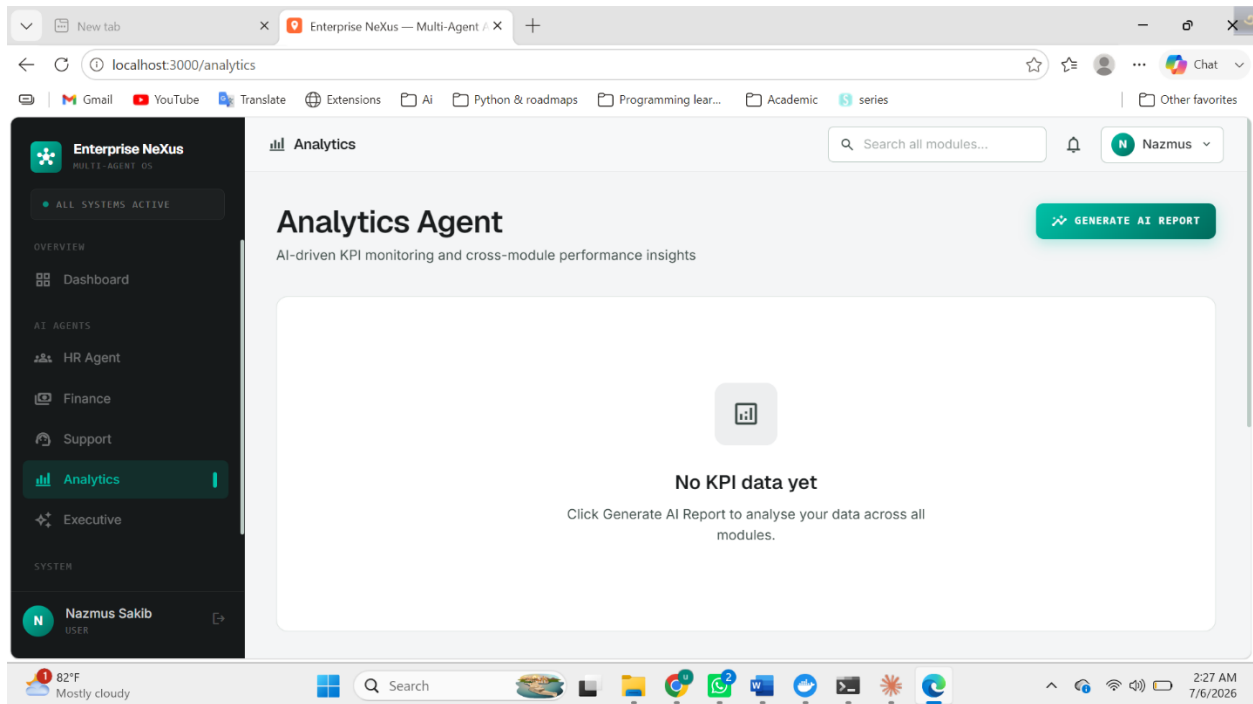


Figure 11: Analytics Agents

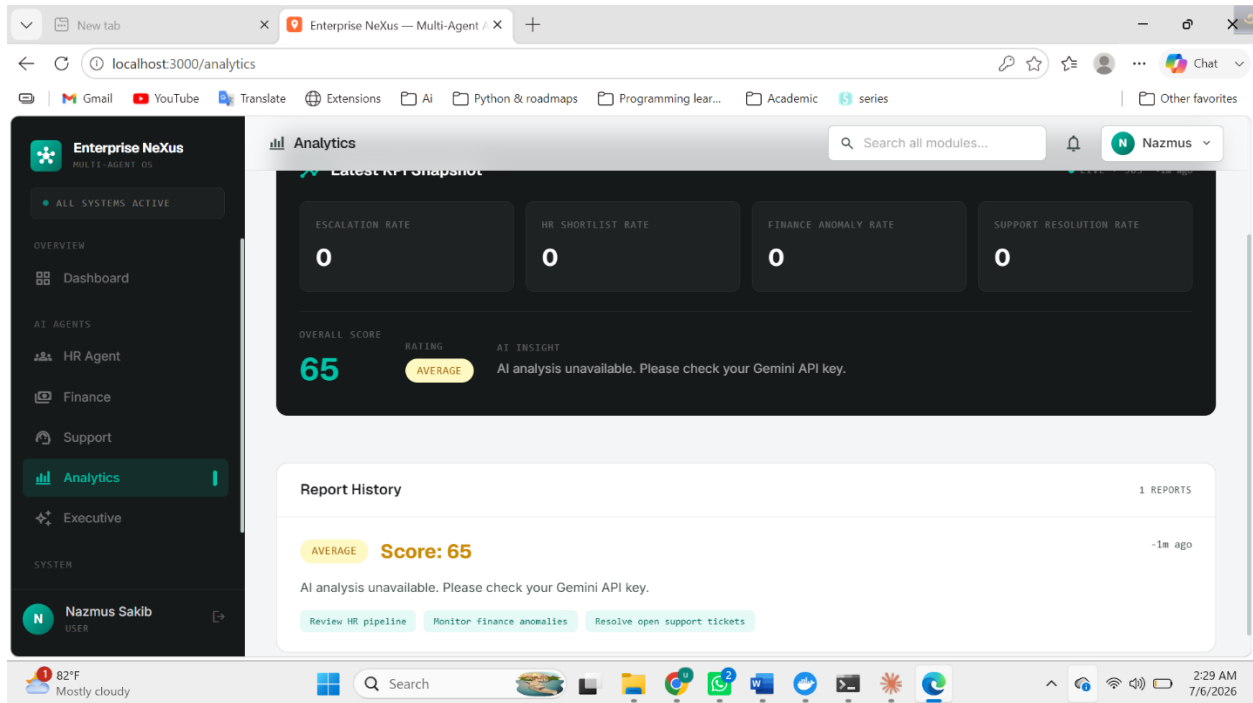


Figure 12: Analytic Agents / Analysis score page

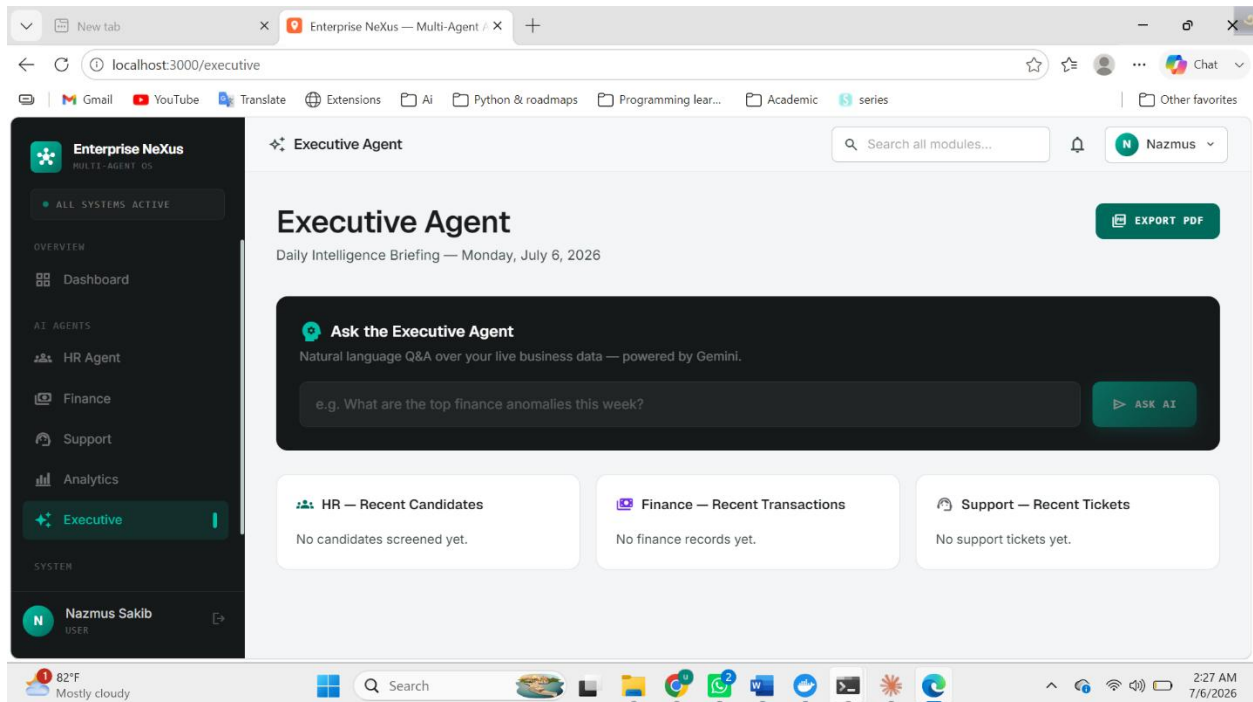


Figure 13: Executive Agents

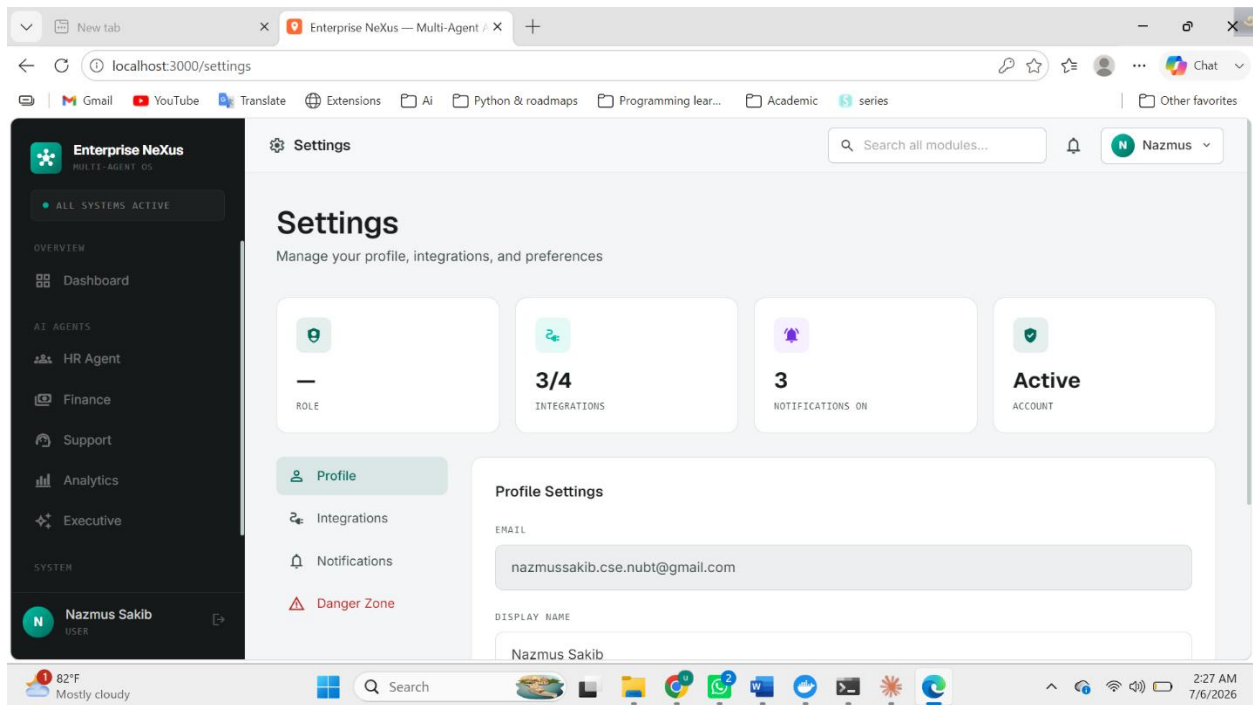


Figure 14: Settings Page

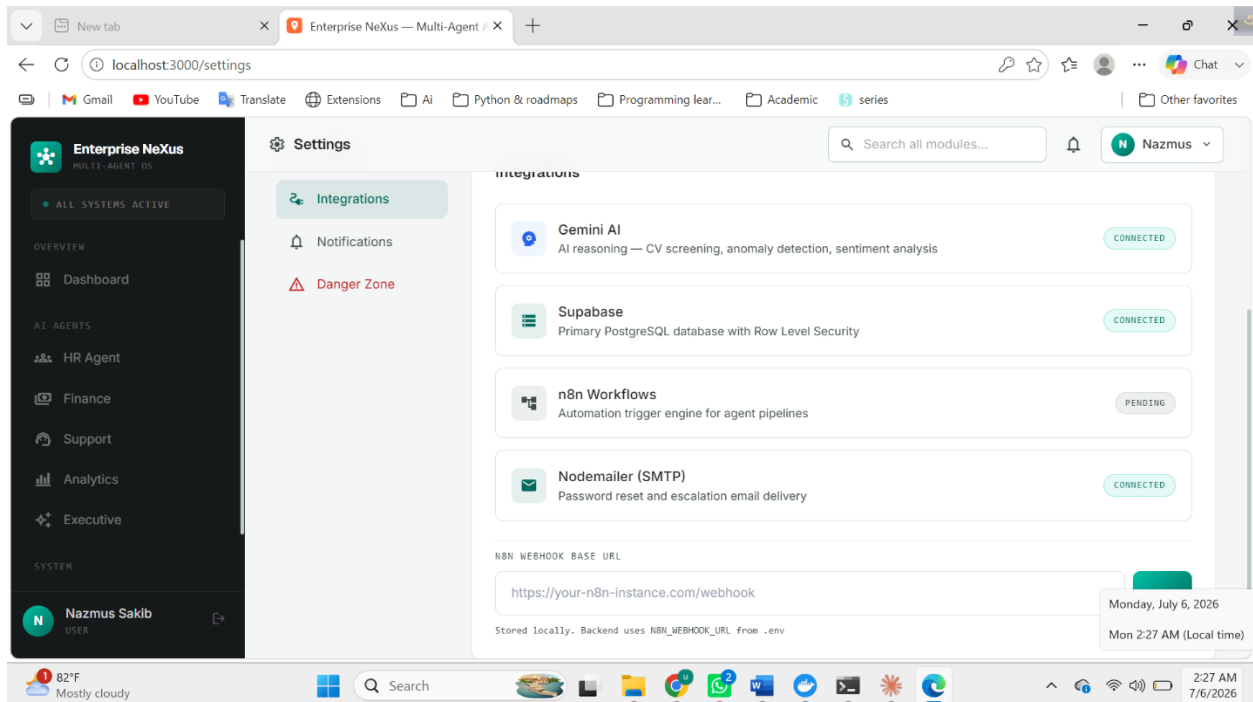


Figure 15: Integration Page

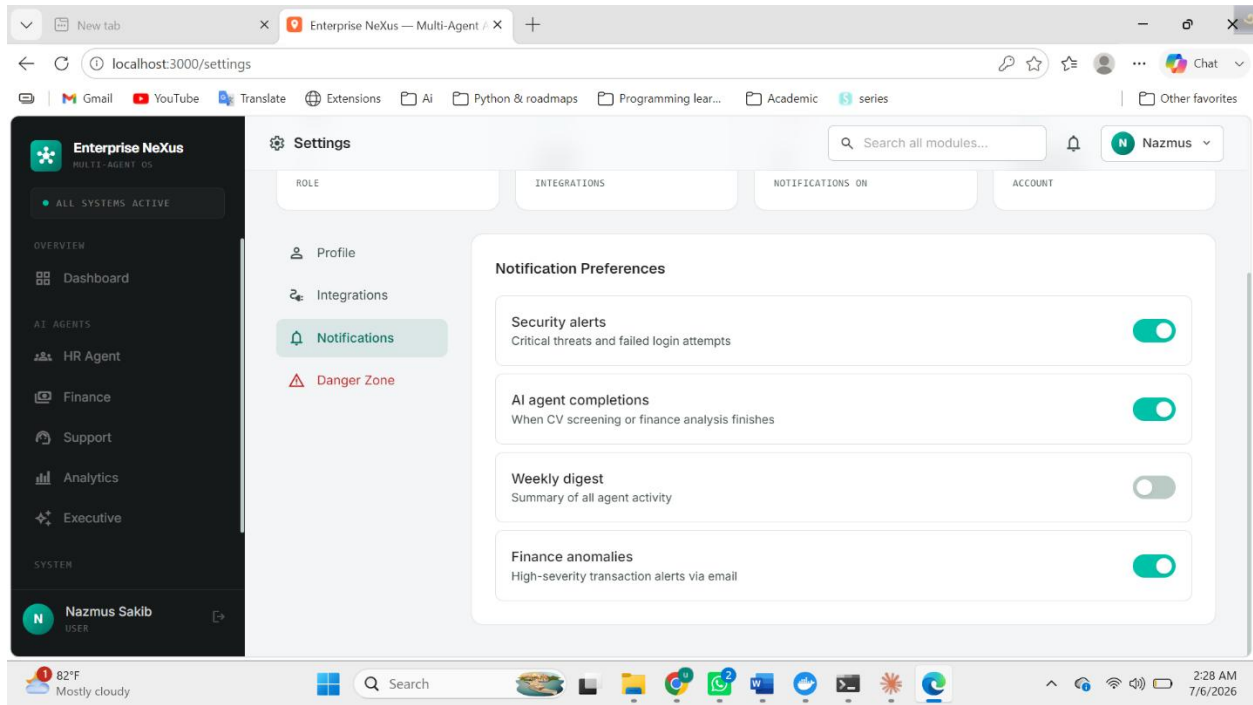


Figure 16: Notification Modification page

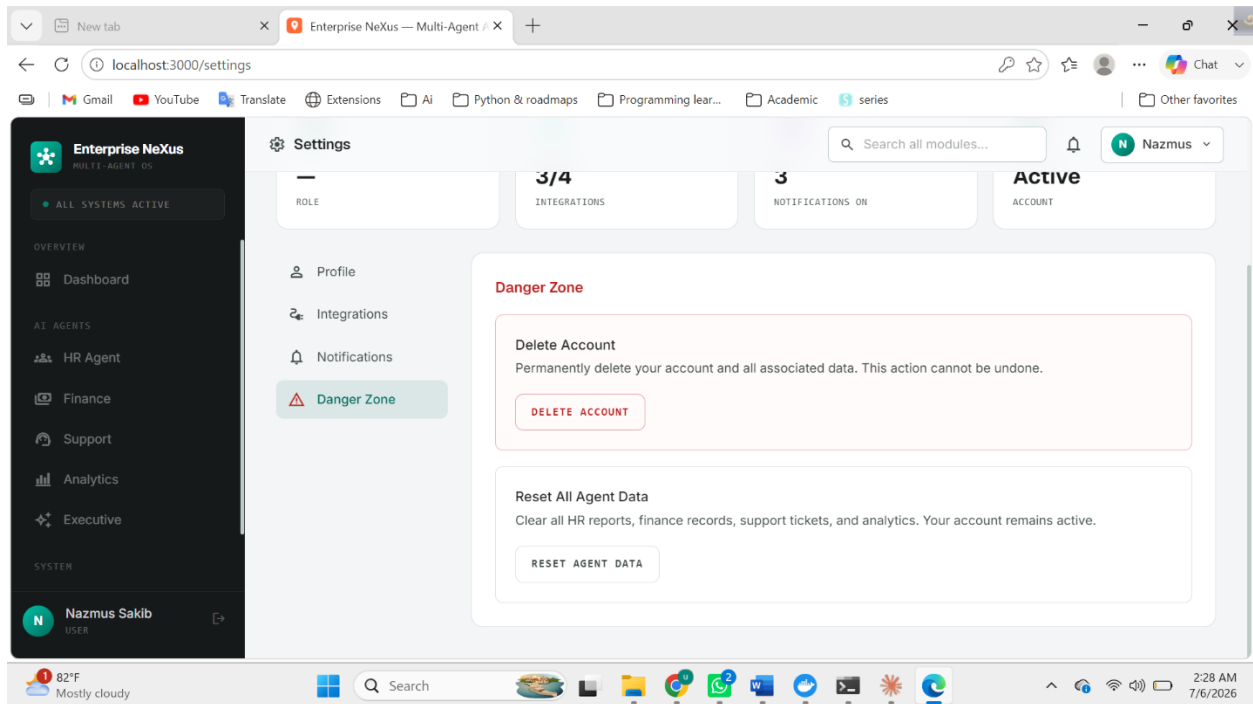


Figure 17: Setting / Danger Zone module